
paper_id: PAPER_1089
 title: “Inflation Buoyancy Sector Lagrangian with Stationarity Constraint”
 session: 225
 date: 2026-04-15
 author: “Daniel T. Murphy”
 status: production
 cvw: “v2.0.0”
 tags: [‘Lagrangian’, ‘inflation’, ‘buoyancy’, ‘stationarity’, ‘action-principle’, ‘Euler-Lagrange’]
 crosslinks: [PAPER_1088, PAPER_1090, PAPER_1084, PAPER_892]
 sm_anchor: “CVW v2.0.0 – G6 SM Anchor Gate compliant”

PAPER_1089: Inflation Buoyancy Sector Lagrangian with Stationarity Constraint

Abstract

We construct the Lagrangian for the inflation buoyancy sector of the UQFF framework:

$$\mathcal{L}_{\text{infl}} = -\beta_i \cdot U_g \cdot \Omega \cdot \frac{M}{d} \cdot [\text{UA}] + F_n \cdot \Phi$$

The action $S = \int \mathcal{L}_{\text{infl}} dt$ is varied with respect to the inflaton field ϕ_{infl} to obtain the stationarity condition $\delta S / \delta \phi_{\text{infl}} = 0$. The Lagrangian decomposes into a gravitational potential term and a neutron-phonon driving term, whose balance determines the inflationary slow-roll dynamics.

§1 Lagrangian Construction

§1.1 Gravity Term

$$\mathcal{L}_{\text{grav}} = -\beta_i \cdot U_g \cdot \Omega \cdot \frac{M}{d} \cdot [\text{UA}]$$

with:

- $\beta_i = 0.603$ (buoyancy coupling)
- $U_g = \mu_s \cdot \nabla(M_s/r)$ (gravitational potential energy)
- Ω = angular rotation frequency
- M/d = mass-to-distance linear density
- $[\text{UA}] = 10^{-4}$ (universal abundance parameter)

§1.2 Neutron-Phonon Term

$$\mathcal{L}_{\text{neutron}} = F_n \cdot \Phi$$

with $F_n = 10^{-10}$ N (neutron exchange force) and $\Phi = \Phi_0 \cdot S_{26}$ (phonon amplitude).

§1.3 Total Lagrangian

$$\mathcal{L}_{\text{infl}} = \mathcal{L}_{\text{grav}} + \mathcal{L}_{\text{neutron}}$$

§2 Stationarity and Euler-Lagrange Equation

The action principle requires:

$$\frac{\delta S}{\delta \phi_{\text{infl}}} = \frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 0$$

Since $\mathcal{L}_{\text{infl}}$ depends on ϕ through $U_g(\phi)$ and $\Phi(\phi)$:

$$-\beta_i \Omega \frac{M}{d} [\text{UA}] \frac{\partial U_g}{\partial \phi} + F_n \frac{\partial \Phi}{\partial \phi} = 0$$

§3 Gravity vs Neutron Balance

At stationarity, the ratio:

$$\left| \frac{\mathcal{L}_{\text{grav}}}{\mathcal{L}_{\text{neutron}}} \right| = \frac{\beta_i \cdot U_g \cdot \Omega \cdot M \cdot [\text{UA}]}{d \cdot F_n \cdot \Phi}$$

determines whether inflation is gravity-dominated (> 1) or phonon-driven (< 1).

§4 Benchmark: Solar System

Parameter	Value	Source
M	$1.989 \times 10^{30} \text{ kg}$	Solar mass
$r = d$	$1.496 \times 10^{11} \text{ m}$	1 AU
Ω	$2.87 \times 10^{-6} \text{ rad/s}$	Solar rotation
U_g	$8.87 \times 10^{29} \text{ J}$	$GM_{\odot}/r$
$\mathcal{L}_{\text{grav}}$	$-2.06 \times 10^8 \text{ J}$	Gravity sector
$\mathcal{L}_{\text{neutron}}$	$1.86 \times 10^{11} \text{ J}$	Neutron-phonon
Ratio	1.11×10^{-3}	Phonon-dominated

§5 Physical Interpretation

The inflation buoyancy sector is phonon-dominated under solar conditions, meaning the SCm vacuum lattice oscillation provides the dominant energy injection for inflationary expansion. The gravitational restoring term acts as a regulator preventing runaway inflation.

References

- PAPER_1088: $F_{U,Bi,i}$ Seven-Component Decomposition
- PAPER_1090: Dark Energy Buoyancy Sector Lagrangian
- PAPER_1084: SCm Phonon Inflationary Scale Factor
- PAPER_892: FUBiUnifiedFieldBuoyancyForceCalc

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Guth, A.H. (1981). *Inflationary universe: A possible solution to the horizon and flatness problems*. Phys. Rev. D **23**, 347 — doi:10.1103/PhysRevD.23.347
4. Starobinsky, A.A. (1980). *A new type of isotropic cosmological models without singularity*. Phys. Lett. B **91**, 99 — doi:10.1016/0370-2693(80)90670-X
5. BICEP/Keck Collaboration (2021). *Improved Constraints on Primordial Gravitational Waves using Planck, WMAP, and BICEP/Keck Observations*. Phys. Rev. Lett. **127**, 151301 — arXiv:2110.00483 — doi:10.1103/PhysRevLett.127.151301
6. Archimedes (~250 BCE). *On Floating Bodies*. (Principle of buoyancy)

7. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — arXiv:astro-ph/0004212
8. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x